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"""
CONN Capacity Test for Larger N (N=256 and N=512)
Same methodology as previous runs, but optimized for bigger networks.
"""

import numpy as np

def generate_patterns(M, N, seed):
    np.random.seed(seed)
    return np.random.choice([0, np.pi], size=(M, N))

def bitflip_plus_jitter(pattern, noise_level, jitter_std, rng):
    noisy = pattern.copy()
    mask = rng.random(len(pattern)) < noise_level
    noisy[mask] = (noisy[mask] + np.pi) % (2 * np.pi)
    noisy = noisy + rng.normal(0, jitter_std, len(pattern))
    return np.mod(noisy, 2 * np.pi)

def compute_overlap(retrieved, target):
    r = np.where(np.cos(retrieved) >= 0, 0.0, np.pi)
    t = np.where(np.cos(target) >= 0, 0.0, np.pi)
    return np.sum(np.abs(np.cos(r) - np.cos(t)) < 0.5) / len(target) *
100

def hebbian_weights(patterns):
    xi = np.cos(patterns)
    J = (xi.T @ xi) / patterns.shape[1]
    np.fill_diagonal(J, 0)
    return J

def hopfield_recall(J, noisy, max_steps=200):
    s = np.cos(noisy).copy()
    for _ in range(max_steps):
        s_new = np.sign(J @ s)
        s_new[s_new == 0] = 1
        if np.array_equal(s_new, s):
            break
        s = s_new
    return np.where(s > 0, 0.0, np.pi)

def conn_annealed(J, noisy, lam, eta_phi, steps):
    N = len(noisy)
    phi = noisy.copy()
    A = np.ones(N)
    for step in range(steps):
        t = step / steps
        lam_t = lam * t
        s, c = np.sin(phi), np.cos(phi)
        coupling = (J @ (A * s)) * c - (J @ (A * c)) * s
        coherence = -2 * lam_t * A**2 * s * c

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    dphi = A * coupling + coherence
    phi = (phi + eta_phi * dphi) % (2 * np.pi)
    dA = -2 * lam_t * A * s**2 - 1.0 * (A - 1)
    A = np.clip(A + 0.03 * dA, 0.01, 2.0)
    return phi

#
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# MAIN EXPERIMENT - Larger Networks
#
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print("=" * 90)
print("CONN vs Hopfield Capacity Test - Larger N (256 & 512)")
print("Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5,
noise=0.30")
print("=" * 90)

noise = 0.30
jitter = 0.5
trials = 30          # Reduced from 50 for speed at large N
test_patterns_per_trial = 3

for N in [256, 512]:
    print(f"\n{'='*80}")
    print(f"                                N = {N}")
    print(f"{'='*80}")
    print(f"  {'M':>4s} {'Hopfield':>10s} {'CONN':>8s} {' $\Delta$ ':>8s}
{'hop_std':>8s} {'conn_std':>8s}")
    print(f"  {'-'*62}")

    hop_cap = 0
    conn_cap = 0
    max_M = int(0.35 * N)    # Reasonable upper bound based on previous
trends

    for M in range(1, max_M + 1):
        hop_ov = []
        conn_ov = []

        for trial in range(trials):
            patterns = generate_patterns(M, N, seed=trial)
            J = hebbian_weights(patterns)

            for mu in range(min(M, test_patterns_per_trial)):
                rng = np.random.RandomState(trial * 10000 + mu)
                noisy = bitflip_plus_jitter(patterns[mu], noise,
jitter, rng)

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ret_h = hopfield_recall(J, noisy)
hop_ov.append(compute_overlap(ret_h, patterns[mu]))

ret_c = conn_annealed(J, noisy, 0.5, 0.02, 400)
conn_ov.append(compute_overlap(ret_c, patterns[mu]))

h_mean = np.mean(hop_ov)
c_mean = np.mean(conn_ov)
h_std = np.std(hop_ov)
c_std = np.std(conn_ov)

if h_mean >= 80: hop_cap = M
if c_mean >= 80: conn_cap = M

# Print strategically
if (M <= 5 or M % max(4, N//32) == 0 or
    abs(h_mean - 80) < 10 or abs(c_mean - 80) < 10):
    print(f"    {M:4d} {h_mean:9.1f}% {c_mean:7.1f}% {c_mean-
h_mean:+7.1f}% {h_std:7.1f}% {c_std:7.1f}%")

# Early stopping if both are clearly below threshold
if h_mean < 60 and c_mean < 60 and M > 20:
    break

ratio = conn_cap / hop_cap if hop_cap > 0 else 0
print(f"\n  _____")
print(f" | Hopfield capacity: M={hop_cap:3d} ( $\alpha$ ={hop_cap/N:.3f})
|")
print(f" | CONN capacity:      M={conn_cap:3d} ( $\alpha$ ={conn_cap/N:.3f})
|")
print(f" | Improvement ratio: {ratio:.2f}×
|")
print(f"  _____")

print("\n" + "="*90)
print("Run completed. Check if the improvement ratio continues to
increase.")

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## RESULTS OF THE RUN IN COLAB

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CONN vs Hopfield Capacity Test - Larger N (256 & 512)
Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5, noise=0.30
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N = 256

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| M     | Hopfield | CONN  | $\Delta$ | hop_std | conn_std |
|-------|----------|-------|----------|---------|----------|
| ----- |          |       |          |         |          |
| 1     | 100.0%   | 89.5% | -10.5%   | 0.0%    | 5.9%     |
| 2     | 100.0%   | 92.0% | -8.0%    | 0.0%    | 4.5%     |
| 3     | 100.0%   | 92.2% | -7.8%    | 0.0%    | 4.1%     |
| 4     | 100.0%   | 92.2% | -7.8%    | 0.0%    | 4.2%     |
| 5     | 100.0%   | 92.6% | -7.4%    | 0.0%    | 3.9%     |
| 8     | 100.0%   | 92.3% | -7.7%    | 0.0%    | 3.8%     |
| 16    | 99.5%    | 90.4% | -9.1%    | 3.7%    | 4.3%     |
| 19    | 99.3%    | 89.5% | -9.8%    | 4.4%    | 4.6%     |
| 20    | 98.7%    | 89.0% | -9.7%    | 6.1%    | 4.8%     |
| 21    | 97.2%    | 88.6% | -8.6%    | 8.5%    | 4.9%     |
| 22    | 96.7%    | 88.4% | -8.3%    | 9.3%    | 5.0%     |
| 23    | 96.4%    | 87.9% | -8.5%    | 9.2%    | 5.0%     |
| 24    | 95.8%    | 87.5% | -8.3%    | 10.0%   | 5.1%     |
| 25    | 95.2%    | 87.2% | -7.9%    | 10.2%   | 5.1%     |
| 26    | 93.1%    | 87.1% | -6.1%    | 12.2%   | 5.2%     |
| 27    | 92.7%    | 86.8% | -5.9%    | 11.7%   | 5.2%     |
| 28    | 91.9%    | 86.4% | -5.5%    | 12.9%   | 5.2%     |
| 29    | 88.8%    | 85.9% | -2.9%    | 14.7%   | 5.4%     |
| 30    | 86.9%    | 85.7% | -1.2%    | 14.8%   | 5.2%     |
| 31    | 86.2%    | 85.5% | -0.8%    | 14.3%   | 5.2%     |
| 32    | 84.8%    | 85.2% | +0.4%    | 15.1%   | 5.1%     |
| 33    | 83.3%    | 85.1% | +1.8%    | 14.3%   | 5.0%     |
| 34    | 81.7%    | 84.9% | +3.2%    | 14.5%   | 5.1%     |
| 35    | 80.1%    | 84.5% | +4.3%    | 13.3%   | 5.3%     |
| 36    | 78.8%    | 83.8% | +5.0%    | 14.2%   | 5.3%     |
| 37    | 77.9%    | 83.8% | +5.9%    | 13.8%   | 5.3%     |
| 38    | 75.7%    | 83.3% | +7.6%    | 13.2%   | 5.6%     |
| 39    | 75.9%    | 82.9% | +7.0%    | 12.9%   | 5.5%     |
| 40    | 74.9%    | 82.8% | +7.9%    | 13.4%   | 5.4%     |
| 41    | 74.1%    | 82.4% | +8.3%    | 12.0%   | 5.4%     |
| 42    | 73.8%    | 82.1% | +8.3%    | 12.1%   | 5.1%     |
| 43    | 73.1%    | 81.8% | +8.6%    | 10.8%   | 5.3%     |
| 44    | 72.0%    | 81.5% | +9.5%    | 10.2%   | 5.4%     |
| 45    | 71.2%    | 81.4% | +10.3%   | 8.7%    | 5.4%     |
| 46    | 70.4%    | 81.1% | +10.7%   | 9.6%    | 5.5%     |
| 47    | 70.8%    | 80.7% | +9.9%    | 9.9%    | 5.5%     |
| 48    | 70.1%    | 80.3% | +10.2%   | 9.9%    | 5.6%     |
| 49    | 70.9%    | 80.2% | +9.3%    | 9.5%    | 5.7%     |
| 50    | 70.5%    | 79.9% | +9.4%    | 9.3%    | 5.3%     |
| 51    | 68.6%    | 79.6% | +11.0%   | 9.5%    | 5.4%     |
| 52    | 69.2%    | 79.3% | +10.2%   | 8.2%    | 5.3%     |
| 53    | 69.0%    | 79.4% | +10.4%   | 7.6%    | 5.5%     |
| 54    | 67.3%    | 79.3% | +11.9%   | 8.5%    | 5.3%     |
| 55    | 67.5%    | 79.1% | +11.6%   | 7.9%    | 5.3%     |

|    |       |       |        |      |      |
|----|-------|-------|--------|------|------|
| 56 | 68.0% | 79.1% | +11.2% | 8.1% | 5.1% |
| 57 | 67.1% | 78.9% | +11.8% | 7.2% | 5.2% |
| 58 | 67.8% | 78.9% | +11.1% | 7.9% | 5.3% |
| 59 | 66.4% | 78.3% | +11.9% | 7.2% | 5.3% |
| 60 | 67.0% | 78.3% | +11.3% | 6.6% | 5.4% |
| 61 | 67.4% | 78.2% | +10.8% | 7.1% | 5.6% |
| 62 | 67.7% | 78.0% | +10.3% | 7.5% | 5.4% |
| 63 | 67.4% | 77.8% | +10.4% | 8.0% | 5.4% |
| 64 | 66.9% | 77.4% | +10.5% | 7.8% | 5.5% |
| 65 | 67.0% | 77.2% | +10.2% | 7.9% | 5.4% |
| 66 | 66.3% | 76.7% | +10.3% | 7.5% | 5.4% |
| 67 | 66.0% | 76.5% | +10.4% | 7.0% | 5.4% |
| 68 | 66.1% | 76.3% | +10.2% | 7.0% | 5.5% |
| 69 | 66.0% | 76.3% | +10.3% | 7.4% | 5.5% |
| 70 | 66.0% | 76.1% | +10.1% | 8.0% | 5.6% |
| 71 | 65.4% | 75.7% | +10.2% | 7.9% | 5.4% |
| 72 | 64.8% | 75.2% | +10.4% | 7.6% | 5.4% |
| 73 | 65.8% | 75.3% | +9.5%  | 7.3% | 5.4% |
| 74 | 65.2% | 74.9% | +9.8%  | 7.1% | 5.5% |
| 75 | 64.7% | 75.0% | +10.3% | 6.8% | 5.5% |
| 76 | 64.5% | 74.5% | +10.0% | 6.7% | 5.5% |
| 77 | 65.3% | 74.4% | +9.1%  | 6.9% | 5.6% |
| 78 | 64.7% | 74.1% | +9.4%  | 6.2% | 5.6% |
| 79 | 64.6% | 73.9% | +9.3%  | 6.0% | 5.5% |
| 80 | 63.9% | 73.9% | +10.0% | 6.8% | 5.6% |
| 81 | 64.2% | 73.6% | +9.5%  | 6.0% | 5.6% |
| 82 | 64.3% | 73.3% | +9.0%  | 6.4% | 5.7% |
| 83 | 63.9% | 73.3% | +9.4%  | 6.2% | 5.6% |
| 84 | 64.4% | 73.4% | +9.0%  | 6.5% | 5.6% |
| 85 | 64.3% | 73.1% | +8.9%  | 6.2% | 5.6% |
| 86 | 63.6% | 72.9% | +9.2%  | 6.4% | 5.5% |
| 87 | 63.4% | 72.7% | +9.3%  | 5.8% | 5.8% |
| 88 | 64.3% | 72.8% | +8.5%  | 6.3% | 5.6% |
| 89 | 62.8% | 72.9% | +10.1% | 6.4% | 5.4% |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| Hopfield capacity: M= 35 ( $\alpha=0.137$ )<br>CONN capacity: M= 49 ( $\alpha=0.191$ )<br>Improvement ratio: 1.40× |
|--------------------------------------------------------------------------------------------------------------------|

N = 512

| M  | Hopfield | CONN  | $\Delta$ | hop_std | conn_std |
|----|----------|-------|----------|---------|----------|
| 1  | 100.0%   | 91.5% | -8.5%    | 0.0%    | 3.2%     |
| 2  | 100.0%   | 91.4% | -8.6%    | 0.0%    | 3.6%     |
| 3  | 100.0%   | 91.9% | -8.1%    | 0.0%    | 3.3%     |
| 4  | 100.0%   | 92.1% | -7.9%    | 0.0%    | 3.2%     |
| 5  | 100.0%   | 92.1% | -7.9%    | 0.0%    | 3.2%     |
| 16 | 100.0%   | 92.2% | -7.8%    | 0.0%    | 2.7%     |
| 32 | 100.0%   | 89.9% | -10.1%   | 0.1%    | 3.0%     |
| 33 | 100.0%   | 89.6% | -10.4%   | 0.1%    | 3.1%     |
| 34 | 100.0%   | 89.5% | -10.5%   | 0.1%    | 3.2%     |
| 35 | 100.0%   | 89.1% | -10.9%   | 0.0%    | 3.3%     |
| 36 | 99.8%    | 89.1% | -10.7%   | 2.1%    | 3.5%     |
| 37 | 99.6%    | 89.0% | -10.6%   | 3.4%    | 3.5%     |
| 38 | 99.8%    | 88.9% | -10.9%   | 1.3%    | 3.4%     |
| 39 | 99.6%    | 88.9% | -10.7%   | 2.7%    | 3.4%     |
| 40 | 99.6%    | 88.7% | -10.9%   | 2.4%    | 3.5%     |

|     |       |       |        |       |      |
|-----|-------|-------|--------|-------|------|
| 41  | 99.5% | 88.6% | -10.9% | 2.0%  | 3.4% |
| 42  | 99.3% | 88.5% | -10.8% | 3.0%  | 3.5% |
| 43  | 99.4% | 88.4% | -11.0% | 3.5%  | 3.6% |
| 44  | 99.4% | 88.1% | -11.3% | 2.5%  | 3.6% |
| 45  | 98.8% | 88.1% | -10.7% | 4.8%  | 3.6% |
| 46  | 99.2% | 87.7% | -11.4% | 3.2%  | 3.6% |
| 47  | 97.1% | 87.5% | -9.7%  | 7.6%  | 3.8% |
| 48  | 95.9% | 87.2% | -8.7%  | 9.3%  | 3.9% |
| 49  | 95.0% | 86.9% | -8.1%  | 10.6% | 3.9% |
| 50  | 93.7% | 86.9% | -6.8%  | 11.4% | 3.9% |
| 51  | 94.4% | 86.7% | -7.7%  | 10.5% | 4.0% |
| 52  | 93.9% | 86.6% | -7.3%  | 11.1% | 4.1% |
| 53  | 92.1% | 86.3% | -5.8%  | 12.7% | 4.1% |
| 54  | 91.7% | 86.2% | -5.5%  | 12.8% | 4.0% |
| 55  | 90.7% | 86.0% | -4.8%  | 13.3% | 4.1% |
| 56  | 89.5% | 85.8% | -3.6%  | 13.8% | 4.1% |
| 57  | 88.6% | 85.6% | -3.0%  | 13.9% | 4.3% |
| 58  | 88.5% | 85.6% | -2.9%  | 13.6% | 4.1% |
| 59  | 87.9% | 85.4% | -2.5%  | 13.9% | 4.2% |
| 60  | 86.0% | 85.4% | -0.6%  | 14.4% | 4.1% |
| 61  | 84.9% | 85.2% | +0.3%  | 14.7% | 4.2% |
| 62  | 83.7% | 84.9% | +1.2%  | 14.9% | 4.3% |
| 63  | 83.4% | 84.8% | +1.4%  | 14.4% | 4.3% |
| 64  | 82.6% | 84.7% | +2.1%  | 14.1% | 4.3% |
| 65  | 80.6% | 84.5% | +3.9%  | 13.7% | 4.4% |
| 66  | 79.6% | 84.5% | +4.9%  | 14.4% | 4.5% |
| 67  | 79.1% | 84.1% | +5.0%  | 14.2% | 4.6% |
| 68  | 76.1% | 84.0% | +7.9%  | 13.4% | 4.5% |
| 69  | 74.6% | 83.7% | +9.1%  | 11.5% | 4.4% |
| 70  | 74.1% | 83.6% | +9.5%  | 11.3% | 4.4% |
| 71  | 73.9% | 83.3% | +9.5%  | 11.4% | 4.5% |
| 72  | 73.1% | 83.2% | +10.2% | 11.2% | 4.4% |
| 73  | 72.8% | 83.1% | +10.3% | 11.2% | 4.4% |
| 74  | 72.1% | 82.9% | +10.8% | 10.0% | 4.4% |
| 75  | 70.8% | 82.9% | +12.1% | 10.3% | 4.5% |
| 76  | 70.8% | 82.9% | +12.1% | 8.7%  | 4.4% |
| 77  | 71.3% | 82.8% | +11.5% | 9.7%  | 4.4% |
| 78  | 70.5% | 82.4% | +11.9% | 8.6%  | 4.4% |
| 79  | 69.9% | 82.3% | +12.4% | 9.0%  | 4.4% |
| 80  | 68.5% | 82.0% | +13.5% | 9.0%  | 4.5% |
| 81  | 68.6% | 81.8% | +13.2% | 8.1%  | 4.5% |
| 82  | 67.8% | 81.8% | +14.0% | 7.5%  | 4.4% |
| 83  | 67.8% | 81.7% | +13.8% | 7.5%  | 4.6% |
| 84  | 67.9% | 81.6% | +13.7% | 6.4%  | 4.5% |
| 85  | 67.7% | 81.4% | +13.7% | 7.6%  | 4.6% |
| 86  | 67.5% | 81.3% | +13.8% | 7.5%  | 4.5% |
| 87  | 67.2% | 81.2% | +14.0% | 7.0%  | 4.5% |
| 88  | 67.0% | 81.2% | +14.2% | 6.8%  | 4.6% |
| 89  | 66.8% | 81.0% | +14.3% | 6.0%  | 4.6% |
| 90  | 66.8% | 80.6% | +13.8% | 6.4%  | 4.6% |
| 91  | 66.3% | 80.5% | +14.2% | 6.8%  | 4.5% |
| 92  | 66.4% | 80.1% | +13.7% | 6.6%  | 4.4% |
| 93  | 67.0% | 80.2% | +13.1% | 6.3%  | 4.4% |
| 94  | 66.3% | 80.1% | +13.7% | 6.3%  | 4.4% |
| 95  | 66.4% | 79.9% | +13.4% | 7.1%  | 4.4% |
| 96  | 65.8% | 79.7% | +13.9% | 6.2%  | 4.5% |
| 97  | 66.1% | 79.6% | +13.6% | 6.3%  | 4.5% |
| 98  | 65.8% | 79.7% | +13.9% | 5.4%  | 4.6% |
| 99  | 65.1% | 79.7% | +14.6% | 6.5%  | 4.6% |
| 100 | 64.9% | 79.3% | +14.4% | 5.3%  | 4.4% |
| 101 | 64.4% | 79.2% | +14.8% | 5.8%  | 4.3% |
| 102 | 64.5% | 79.2% | +14.7% | 5.6%  | 4.3% |
| 103 | 64.9% | 78.9% | +14.0% | 5.7%  | 4.3% |

|     |       |       |        |      |      |
|-----|-------|-------|--------|------|------|
| 104 | 64.6% | 78.9% | +14.3% | 5.2% | 4.2% |
| 105 | 65.1% | 78.7% | +13.5% | 5.2% | 4.4% |
| 106 | 64.6% | 78.6% | +14.0% | 5.3% | 4.2% |
| 107 | 64.8% | 78.6% | +13.8% | 5.6% | 4.3% |
| 108 | 64.0% | 78.4% | +14.4% | 5.9% | 4.4% |
| 109 | 64.4% | 78.3% | +13.9% | 5.4% | 4.3% |
| 110 | 64.2% | 78.3% | +14.1% | 5.8% | 4.2% |
| 111 | 64.5% | 78.3% | +13.8% | 5.1% | 4.3% |
| 112 | 64.6% | 78.1% | +13.5% | 5.8% | 4.3% |
| 113 | 64.8% | 77.8% | +13.0% | 5.2% | 4.2% |
| 114 | 64.0% | 77.6% | +13.7% | 5.9% | 4.3% |
| 115 | 63.4% | 77.6% | +14.1% | 5.6% | 4.4% |
| 116 | 63.7% | 77.5% | +13.7% | 5.1% | 4.4% |
| 117 | 64.0% | 77.3% | +13.3% | 5.7% | 4.3% |
| 118 | 63.6% | 77.2% | +13.6% | 5.5% | 4.1% |
| 119 | 64.0% | 76.9% | +12.9% | 5.3% | 4.1% |
| 120 | 63.9% | 77.0% | +13.1% | 5.1% | 4.1% |
| 121 | 63.4% | 76.9% | +13.5% | 5.5% | 4.2% |
| 122 | 63.7% | 76.8% | +13.1% | 5.6% | 4.4% |
| 123 | 64.3% | 76.7% | +12.5% | 5.6% | 4.4% |
| 124 | 63.8% | 76.7% | +13.0% | 5.2% | 4.3% |
| 125 | 63.3% | 76.5% | +13.2% | 5.7% | 4.2% |
| 126 | 63.5% | 76.4% | +12.9% | 5.3% | 4.2% |
| 127 | 63.1% | 76.4% | +13.3% | 5.8% | 4.1% |
| 128 | 63.1% | 76.2% | +13.2% | 5.4% | 4.1% |
| 129 | 62.8% | 76.1% | +13.3% | 5.2% | 4.1% |
| 130 | 62.8% | 76.0% | +13.2% | 5.7% | 4.1% |
| 131 | 62.7% | 76.1% | +13.4% | 5.8% | 4.1% |
| 132 | 62.8% | 76.0% | +13.2% | 5.6% | 4.1% |
| 133 | 63.2% | 75.7% | +12.6% | 5.1% | 4.2% |
| 134 | 63.4% | 75.9% | +12.5% | 4.9% | 4.0% |
| 135 | 63.1% | 75.8% | +12.7% | 5.6% | 4.1% |
| 136 | 63.2% | 75.8% | +12.5% | 4.6% | 3.9% |
| 137 | 63.6% | 75.8% | +12.3% | 5.5% | 3.8% |
| 138 | 62.9% | 75.7% | +12.8% | 5.0% | 3.8% |
| 139 | 62.9% | 75.6% | +12.7% | 4.9% | 3.8% |
| 140 | 63.2% | 75.3% | +12.1% | 5.0% | 3.9% |
| 141 | 63.0% | 75.5% | +12.4% | 4.9% | 3.9% |
| 142 | 62.9% | 75.4% | +12.5% | 5.3% | 3.8% |
| 143 | 61.9% | 75.1% | +13.3% | 4.7% | 3.9% |
| 144 | 62.7% | 75.1% | +12.4% | 5.2% | 3.9% |
| 145 | 62.7% | 75.1% | +12.4% | 4.6% | 3.9% |
| 146 | 62.5% | 74.8% | +12.3% | 5.3% | 3.9% |
| 147 | 62.9% | 74.8% | +11.9% | 5.1% | 3.9% |
| 148 | 63.2% | 74.7% | +11.5% | 4.9% | 4.0% |
| 149 | 62.8% | 74.8% | +12.0% | 4.8% | 3.8% |
| 150 | 63.1% | 74.6% | +11.5% | 4.7% | 3.8% |
| 151 | 62.5% | 74.7% | +12.2% | 4.7% | 3.8% |
| 152 | 62.8% | 74.5% | +11.7% | 4.6% | 3.7% |
| 153 | 62.5% | 74.4% | +11.9% | 4.5% | 3.7% |
| 154 | 62.8% | 74.4% | +11.6% | 4.5% | 3.8% |
| 155 | 62.5% | 74.5% | +11.9% | 5.1% | 3.8% |
| 156 | 62.5% | 74.6% | +12.1% | 4.7% | 3.7% |
| 157 | 62.6% | 74.6% | +12.0% | 4.5% | 3.6% |
| 158 | 62.8% | 74.4% | +11.6% | 5.3% | 3.6% |
| 159 | 62.7% | 74.4% | +11.7% | 5.0% | 3.6% |
| 160 | 63.0% | 74.4% | +11.4% | 4.4% | 3.5% |
| 161 | 62.5% | 74.3% | +11.7% | 4.8% | 3.4% |
| 162 | 62.8% | 74.1% | +11.3% | 4.7% | 3.4% |
| 163 | 62.4% | 74.1% | +11.7% | 4.8% | 3.5% |
| 164 | 62.9% | 74.0% | +11.2% | 4.4% | 3.6% |
| 165 | 62.3% | 73.9% | +11.6% | 4.6% | 3.6% |
| 166 | 62.8% | 73.7% | +10.9% | 4.5% | 3.4% |

|     |       |       |        |      |      |
|-----|-------|-------|--------|------|------|
| 167 | 62.3% | 73.8% | +11.5% | 4.7% | 3.5% |
| 168 | 62.9% | 73.7% | +10.8% | 5.0% | 3.4% |
| 169 | 62.6% | 73.6% | +11.1% | 5.0% | 3.4% |
| 170 | 62.6% | 73.5% | +10.9% | 4.4% | 3.4% |
| 171 | 62.5% | 73.5% | +11.0% | 4.7% | 3.6% |
| 172 | 62.5% | 73.5% | +11.0% | 4.8% | 3.6% |
| 173 | 62.7% | 73.4% | +10.7% | 5.0% | 3.4% |
| 174 | 62.0% | 73.3% | +11.3% | 5.1% | 3.5% |
| 175 | 61.8% | 73.2% | +11.4% | 4.5% | 3.5% |
| 176 | 61.7% | 73.2% | +11.6% | 4.5% | 3.5% |
| 177 | 62.0% | 73.2% | +11.2% | 4.6% | 3.6% |
| 178 | 62.2% | 73.2% | +11.0% | 4.6% | 3.7% |
| 179 | 61.8% | 73.0% | +11.2% | 4.0% | 3.8% |

|                                             |  |
|---------------------------------------------|--|
| Hopfield capacity: M= 65 ( $\alpha=0.127$ ) |  |
| CONN capacity: M= 94 ( $\alpha=0.184$ )     |  |
| Improvement ratio: 1.45×                    |  |

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Run completed. Check if the improvement ratio continues to increase.